

THE HEALING EFFECT: Generating Human Spares



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War veterans return home with all kinds of injuries—some physical, others psychological. A missing limb, an ear blown off, scars of wars can be seen all over. Advances in technology that could fill out missing pieces sound good, do they not?

Researchers at Wake Forest Institute for Regenerative Medicine at Wake Forest Baptist Medical Center, USA, have custom-designed a 3D printer and successfully printed ear, bone and muscle structures to take care of just that. The developed structures have been implanted in animals and matured into functional tissue, apart from developing a system of blood vessels. Probably the future does hold an all-perfect world. Let us take a look at how they do it.

Printing body parts

In a typical plastic modelling 3D printer, basics would involve melting down the material and shaping it back in the form of the required design. However, biological tissue would not be suited to a similar setup and could be easily destroyed. Additionally, a support system would be required in place for the biomaterial to hold onto—similar

to the support system in a traditional 3D printer.

In addition to outputting cells, most bio-printers also output a dissolvable gel to support and protect the cells during printing. Organovo's bio-printing process centres around identification of architectural and compositional elements of a target tissue, and creation of a design that can be utilised by a bio-printer to generate that tissue in the laboratory environment.

Once a tissue design is established, the first step is to develop the bioprocess protocols required to generate multi-cellular building blocks from cells that would be used to build the target tissue.

According to Organovo, the bio-printing process can be tailored to produce tissues in a variety of formats. Ranging from micro-scale tissues contained in standard multi-well tissue culture plates to larger structures suitable for placement onto bioreactors for biomechanical conditioning.

Complex organs are difficult to print.

"The simplest organ is skin, which is a flat structure. As organs move up the ladder of complexity, from tubular structures such as blood vessels, hollow organs like the bladder and stomach, and solid organs like the kidney and liver, these require multiple cell types and more oxygen because these are dense with cells. Being able to meet the oxygen requirements of solid organs is a major challenge," explained the bio-printing team at Wake Forest Institute for Regenerative Medicine.

Materials involved in casting

Scaffold materials involved in designing have to be biocompatible, of suitable strength and have shaping characteristics. The materials being used currently are selected either due to their compatibility with cell growth and function or due to their cross-linking or extrusion charac-

Fig. 1: A bio-printer in action

